

SYSTEM ARCHITECTURE & PRODUCT SPECIFICATION

Gaon Ka Swad

Multi-Outlet Cloud Kitchen Platform

Comprehensive Product Requirements Document (PRD)

End-to-End Business Logic, System Workflows, Role Deep-Dives & Data Schemas

DOCUMENT CONTROL & SYSTEM DETAILS

Document Version:	1.0.0 (Production Release Specification)
Target Stakeholders:	Admin (Owner), Kitchen Outlet Managers, Customers
Core System Engine:	React 19 SPA (Vite) + Express.js + Supabase Realtime DB
Cloud Infrastructure:	PostgreSQL / Supabase + Express Middleware API
Kitchen Hubs Covered:	Bangalore (HSR, Kadabeesanahalli, etc.) & Bhubaneswar
Prepared For:	Cloud Kitchen Executive & Engineering Leadership
Effective Date:	October 2026

Confidential • Internal Enterprise Cloud Kitchen Specification

1. Executive Summary & Evolutionary Journey

Architectural vision and prompt-driven evolutionary synthesis

1.1 Platform Concept & Product Vision

Gaon Ka Swad is a modern, artisanal multi-outlet cloud kitchen operating across multiple metropolitan cities (initially Bangalore and Bhubaneswar). The brand specializes in royal dum biryanis, 24-hour slow-cooked gravies, smoky clay-oven tandoori grills, and authentic regional Indian dishes delivered in sealed eco-handis. Unlike standard single-kitchen food-delivery apps, Gaon Ka Swad is engineered as an enterprise-grade, distributed multi-tenant platform where each outlet functions as an autonomous culinary node while sharing unified master catalog data, centralized accounting, cross-hub loyalty (Swad Coins), and dynamic geofenced delivery zones.

1.2 Prompt-Driven Evolutionary Sprints

The platform was constructed iteratively through sequential, high-precision prompts. The system architecture evolved through the following distinct milestones:

- **Sprint 1 - Artisanal Identity & Customer Storefront:** Established high-fidelity culinary UI, category navigation, dish cards with aroma notes, culinary storytelling, dish variants (e.g. 500g vs 1kg handis), and add-on selectors.
- **Sprint 2 - Multi-Outlet Geofencing & Location Engine:** Introduced 6-digit Indian PIN code auto-resolution against delivery zone arrays, outlet auto-assignment, and the Outlet Switch Safeguard modal to prevent cross-kitchen cart contamination.
- **Sprint 3 - Dual-Layer Master Catalog & Atomic Inventory:** Engineered global catalog synchronization with outlet-level overrides. Implemented atomic database portion decrements via PostgreSQL triggers to prevent overselling popular dum handis.
- **Sprint 4 - Role-Based Access Control (RBAC):** Segregated administrative boundaries into three clear actors: Super Admin (Owner), Outlet Kitchen Managers (scoped strictly to single outlet hubs), and Phone-first Customers.
- **Sprint 5 - Kitchen Display System (KDS) & Order Pipeline:** Designed live manager order board with Web Audio API chime notifications, real-time status transitions (Received -> Confirmed -> Preparing -> Ready -> Delivered/Cancelled), and reason-based cancellation workflows.
- **Sprint 6 - Swad Coins Loyalty Ledger & Dynamic Coupons:** Built a double-entry coin ledger (1 Coin = ₹1) with automated post-delivery reward generation, claim vaults, transaction history, and coupon rules engine with first-order gating.
- **Sprint 7 - Verified Purchase Tastings & Reviews:** Implemented a 7-day post-delivery verified purchase review window with star ratings, tasting feedback, and dynamic catalog rating rollups.
- **Sprint 8 - Executive Dashboard & Exact Order Reconciliation:** Compacted Block 3 into a 1-line top-right date filter (All, Today, 7D, 30D, custom dd/mm/yyyy range) and resolved order count discrepancies by strictly separating Gross Revenue (₹16,737 from 13 active orders) from Cancelled records.

2. Role Specification: Super Admin (Owner)

Centralized operational, financial, catalog, and multi-hub governance

2.1 Executive Dashboard & Revenue Analytics Engine

The Owner Dashboard provides consolidated visibility across all operating cloud kitchen outlets. The system features an ultra-compact "Orders & Revenue" financial nerve center (Block 3) optimized for executive decision-making:

- **Compact One-Liner Date Filter:** Positioned at the top-right of the banner. Features quick presets: All, Today, 7D, 30D, alongside interactive full-click "From Date -> To Date" picker supporting dd/mm/yyyy formatting and instant reset.
- **Gross Revenue Metric:** Calculates valid revenue from active orders: $\text{Gross Revenue} = \text{Sum}(\text{Delivered Orders}) + \text{Sum}(\text{In-Process Orders})$. Excludes cancelled orders to ensure zero phantom earnings.
- **Exact Order Count Reconciliation:** Displays the precise active order count beneath the rupee amount (e.g. ₹6,737 across 13 active orders), reconciling 11 Delivered + 2 In-Process orders while accurately categorizing the 1 Cancelled order separately.
- **4-Column Breakdown:** 1. Revenue (Total valid revenue & count), 2. Delivered (Completed revenue & count), 3. In Process (Active preparation/transit & count), 4. Cancelled (Lost revenue & cancellation count).
- **Average Order Value (AOV):** Prominently presented in the card footer: $\text{AOV} = \text{Gross Revenue} / \text{Valid Orders}$ (e.g. ₹518/order). Fully responsive across mobile, tablet, and desktop.

FINANCIAL CALCULATION FORMULAS

- $\text{Valid Orders} = \text{Count}(\text{Delivered}) + \text{Count}(\text{In-Process})$ [Excludes Cancelled]
- $\text{Gross Revenue} = \text{Subtotal} + \text{Packaging} + \text{DeliveryFee} + \text{Tax} - \text{Discounts}$ (for Valid Orders)
- $\text{Net Realized Revenue} = \text{Delivered Revenue only}$
- $\text{Average Order Value (AOV)} = \text{Gross Revenue} / \text{Valid Orders}$

2.2 Master Catalog & Outlet Override Architecture

The catalog operates on a dual-layer data architecture: a central master product definition and an outlet-specific JSONB configuration array (ProductOutletConfig). The Owner has sole authority to:

- **Global Item Definition:** Configure product titles, Hindi names, slugs, category assignments, detailed culinary descriptions, secret clay-pot cooking methods, aroma profiles, veg/non-veg tags, and Jain-friendly flags.
- **Portion & Inventory Controls:** Set per-outlet portion limits: null (unlimited portions), 0 (marked sold out), or positive integers (finite daily dum portions that atomically decrement as orders are confirmed).
- **Featured & Bestseller Toggles:** Highlight specific regional specialties per outlet independently (e.g., Dum Biryani featured in Bangalore HSR while Dal Makhani featured in Bhubaneswar).
- **Variants & Add-on Bundles:** Define variant pricing (e.g. 500g Handi @ ₹349 vs 1kg Family Pack @ ₹649) and optional add-ons (extra burani raita @ ₹40, boiled egg @ ₹25, salan gravy @ ₹50).

2.3 Multi-Outlet Hubs & Regulatory Governance

- **14-Digit Numeric FSSAI License:** Mandatory regulatory compliance for Indian food safety; validated as strictly numeric 14 digits per outlet.
- **Financial Guardrails:** Define per-outlet Minimum Order Value (MOV, e.g. ₹200), Free Delivery Threshold (e.g. ₹499), and standard Packaging Fee (e.g. ₹25).

- **Custom Brand Storytelling:** Each outlet possesses custom hero fire-lines, trust badge ratings ("4.9 & 2.8k+ Google & Zomato"), and dedicated "About Kitchen" narratives.

2. Super Admin: Promotions, Zones & Loyalty

Geofencing, coupon creation rules, and Swad Coins ledger administration

2.4 Delivery Zones & Dynamic Geofencing

To guarantee food reaches customers at peak temperature within 30-40 minutes, each outlet is mapped to multiple delivery zones:

- **PIN Code Array Mapping:** Each delivery zone encapsulates a dedicated array of 6-digit postal PIN codes. When a customer enters their PIN, the server matches the active zone and binds the session to the target outlet.
- **Custom Zone Pricing:** Allows setting variable delivery fees (e.g. ₹140 for core 3km zone, ₹160 for outer radius) and estimated delivery transit times.

2.5 Enterprise Coupon Rules Engine

The platform includes a robust promotional discounting system managed exclusively by the Super Admin:

- **Discount Mechanisms:** Supports percentage discounts (with mandatory maxDiscountAmount caps, e.g. 20% up to ₹100) or fixed flat rupee discounts.
- **Minimum Order Value (MOV):** Coupons strictly enforce basket subtotal thresholds before applying discounts.
- **Audience & Scope Restrictions:** Can be targeted globally across all outlets or restricted to designated outlets. Supports "First Order Only" gating or "Logged-in Only" restrictions.
- **Usage Limiting:** Enforces both total campaign usage caps and per-customer usage limits tracked by customer phone number in the coupon_redemptions table.

2.6 Swad Coins Loyalty Program Administration

Swad Coins represent a proprietary digital cashback and customer retention engine (1 Swad Coin = ₹1.00 INR):

- **Reward Percentage & Caps:** Standard reward rule: 5% to 10% cashback on eligible subtotal post-delivery, with an enforced maximum cap of 100 coins per order.
- **Admin Credit / Debit Controls:** Admin can manually credit compensation coins or adjust balances with mandatory audit trail descriptions stored in swad_coin_transactions.
- **Batch Dispatch Engine:** Includes automated cron routines and manual "Generate Rewards" trigger (SwadCoinDispatch) scanning delivered orders and issuing claimable reward records.

SWAD COIN TRANSACTION TYPES & AUDIT TRAIL

- **EARN:** Credited to customer upon claiming delivered order reward.
- **REDEEM:** Deducted at checkout when customer spends coins (1 Coin = ₹1).
- **REFUND:** Automatically returned to customer if order is cancelled.
- **ADMIN_CREDIT:** Manual goodwill/support credit by Super Admin.
- **ADMIN_DEBIT:** Manual correction/adjustment by Super Admin.

3. Role Specification: Outlet Kitchen Manager

Kitchen Display System (KDS), order fulfillment, and local stock control

3.1 Scoped Access & Kitchen Operations

Outlet Kitchen Managers are frontline operational leaders. Their access is strictly compartmentalized to their assigned physical cloud kitchen (e.g., `manager.hsr@gaonkaswad.in` is locked exclusively to `blr-hsr`). They do not see financials, coupons, or catalog pricing of other outlets.

3.2 Kitchen Order Display (KDS) & Order Pipeline

The KDS provides real-time order queue management designed for fast-paced commercial kitchens:

- **Live Audio Chime Alert:** Synthesized Web Audio API bell chimes immediately when a new order enters the "Received" state, alerting chefs without requiring screen gazing.
- **Real-Time Supabase Synchronization:** Subscribed to PostgreSQL Realtime events. New orders, status changes, and cancellations update instantly on screen without browser refreshes.
- **Strict Sequential Status Pipeline:** Orders advance through standardized operational phases:

KITCHEN ORDER STATUS STATE MACHINE

1. Received: Order placed by customer, awaiting kitchen confirmation.
2. Confirmed: Kitchen acknowledges order; ticket printed/sent to cook station.
3. Preparing in Kitchen: Handis on dum flame, tandoor roasting active.
4. Ready for Pickup: Sealed in eco-handi bags, awaiting delivery partner pickup.
5. Out for Delivery: Rider assigned and in transit to customer doorstep.
6. Delivered: Handed over to customer; triggers Swad Coin reward generation.
7. Cancelled: Aborted order with mandatory reason logging and automatic rollbacks.

3.3 Order Inspection & Packaging Details

- **Itemized Ticket Snapshots:** Displays line items with selected variants (e.g. 1kg Handi), spice levels (Extra Spicy), and selected add-ons (Extra Salan). Prices are preserved as historical snapshots.
- **Cooking & Gate Instructions:** Displays customer notes (e.g., "Less oil in biryani, ring door bell twice").
- **Eco-Cutlery Indicator:** Prominently flags whether the customer requested wooden disposable cutlery or opted out for zero waste.
- **Delivery Type & Scheduling:** Differentiates "Immediate Delivery" from "Scheduled Delivery" slots with visual color-coded badges to prevent premature cooking.

3.4 Cancellation Handling & Atomic Rollbacks

When an order must be cancelled (e.g. customer request or out-of-stock emergency), the Manager utilizes the Cancel Order modal with predefined reasons. The system immediately executes:

- **Inventory Reversal:** Restores portion counts back to the product outlets config.
- **Loyalty Coin Refund:** If the customer redeemed Swad Coins, an automated REFUND transaction is created, restoring the customer balance.

- **Audit Logging:** Records the cancellation timestamp and reason permanently in public.orders.

4. Role Specification: Customer Experience

Location-first onboarding, artisanal menu browsing, checkout & post-order delight

4.1 Location-First Geocoding & Outlet Safeguard

To eliminate food delivery disappointment, the customer experience begins with location detection:

- **6-Digit PIN Code Auto-Lookup:** Validates Indian postal PIN codes against all outlet delivery zones. Automatically resolves City, State, Area, and assigned Cloud Kitchen Outlet.
- **Outlet Switching Safeguard:** If a customer changes their delivery PIN while holding items from an outlet in their cart, the system intercepts with a modal warning: "Your cart contains items from a different kitchen. Would you like to clear cart and switch, or keep current outlet?".

4.2 Artisanal Menu Browsing & Customization

The storefront showcases culinary heritage with deep sensory details:

- **Dietary & Lifestyle Filters:** One-click toggles for Pure Veg, Non-Veg, Jain-Friendly (prepared without root vegetables/onions/garlic), and Spice Intensity (Mild, Medium, Spicy, Extra Spicy).
- **Sensory Dish Profile:** Every dish details culinary heritage stories, slow-cooking dum techniques, secret spice masalas, aroma notes, portion weights, and allergen disclosures.
- **Interactive Customization:** Modal selector for variant sizing (Handi 500g vs 1000g) and accompaniment add-ons with dynamic price updates.

4.3 Server-Authoritative Cart & Pricing Engine

All financial calculations are computed server-side to prevent client tampering:

- **Cart Subtotal:** Sum of all selected line items (base variant + add-ons) * quantity.
- **Packaging Fee:** Nominal per-order fee for food-grade earthen handis and tamper-proof packaging.
- **Dynamic Delivery Fee:** Standard zone fee applied; automatically waived (₹0) when subtotal meets or exceeds the outlet Free Delivery Threshold (e.g. ₹499).
- **5% Statutory Restaurant GST:** Accurately computed as 5% on the net food subtotal after applicable discounts.
- **Final Payable Total:** $\text{Total} = \text{Max}(0, \text{Subtotal} - \text{Discounts} + \text{PackagingFee} + \text{DeliveryFee} + \text{GST})$.

4.4 Checkout & Phone-First Authentication

- **Phone OTP Verification:** Frictionless authentication via 6-digit phone OTP (SMS/WhatsApp gateway integrated, with beta test mode support for rapid onboarding).
- **Saved Address Book:** Supports multiple delivery addresses labeled as Home, Work, or Other, with default selection and landmark precision.
- **Flexible Delivery Modes:** Supports Immediate Dispatch (30-40 min) or Scheduled Delivery (date picker with 1-hour delivery time slots).
- **Payment Versatility:** Supports Cash on Delivery (COD) and Online Payments (UPI, Cards, Netbanking via Razorpay-ready integration).

4. Customer Loyalty, Vault & Verified Reviews

Retention loops, reward claiming, and authenticated customer reviews

4.5 Swad Coins Rewards Vault & Claim Flow

Unlike passive loyalty points, Swad Coins create an engaging, active claim loop that drives repeated app visits:

- **Pending Reward Generation:** When an order status changes to "Delivered", the system calculates a 5%-10% reward (up to 100 coins) and creates a PENDING reward record.
- **Customer Claim Vault:** The customer navigates to their Swad Coins Vault in their profile and clicks "Claim Reward". This transfers the coins into their live balance and records an EARN transaction.
- **Redemption at Checkout:** 1 Swad Coin = ₹1.00 INR. Customers can toggle "Redeem Swad Coins" at checkout, deducting the exact amount from their total bill.
- **Expiration Window:** Unclaimed rewards feature a 60-90 day expiration window, incentivizing prompt return engagement.

4.6 Verified Purchase Tasting Reviews

To ensure 100% review authenticity and protect brand reputation:

- **Verified Purchase Gating:** Reviews can strictly only be submitted for dishes actually delivered in a completed order. No guest or pre-purchase reviews are allowed.
- **7-Day Review Window:** Customers can review items up to 7 days after the delivered_at timestamp. Once the deadline passes, review eligibility expires.
- **Granular Star Ratings & Comments:** 1 to 5-star rating with optional 500-character tasting commentary.
- **Automatic Catalog Rating Rollup:** Submitted ratings immediately update the product average rating and review counts on the public storefront.

END-TO-END VERIFIED REVIEW LIFECYCLE

1. Order reaches "Delivered" status -> delivered_at timestamp recorded.
2. Endpoint /api/reviews/eligibility verifies customer phone & 7-day window.
3. Customer rates each dish individually in Order History.
4. Reviews published to public.product_reviews; product rating re-indexed.

5. Database Architecture & Technical Specs

PostgreSQL schema, triggers, and Row Level Security (RLS)

5.1 Master Table Inventory

The Supabase PostgreSQL database comprises 14 normalized, highly indexed tables:

- public.outlets:** Physical cloud kitchen hubs, addresses, 14-digit FSSAI IDs, fees, thresholds.
- public.delivery_zones:** Geofenced 6-digit postal PIN code arrays linked to outlets with delivery SLAs.
- public.products:** Master culinary catalog, descriptions, images, variants, allergens, outlet configs.
- public.customers:** Phone-first customer profiles, verification status, welcome discount flags.
- public.customer_addresses:** Customer saved addresses (Home, Work, Other) with full address and landmarks.
- public.orders:** Normalized orders master with status pipeline, financial breakdowns, and timestamps.
- public.order_items:** Line-item snapshots with historical unit prices, selected variants, and add-ons.
- public.coupons:** Promotional discount codes, percentage/fixed values, caps, MOVs, and limits.
- public.coupon_redemptions:** Historical ledger tracking coupon usage per customer phone and order.
- public.swad_coin_rewards:** Post-delivery pending cashback rewards with claim states and expiry dates.
- public.swad_coin_transactions:** Double-entry loyalty ledger (EARN, REDEEM, REFUND, ADMIN_CREDIT, ADMIN_DEBIT).
- public.swad_coin_dispatches:** Audit records of automated and manual reward distribution runs.
- public.product_reviews:** Verified customer tasting reviews tied to orders, items, and ratings.
- public.profiles:** Role-based authorization profiles (owner, outlet_manager, customer) linked to auth.

5.2 Automated Database Triggers & Functions

- **trg_generate_order_id:** Before INSERT on public.orders. Automatically assigns sequential human-readable IDs formatted as GKSWAD-00001, GKSWAD-00002 using PostgreSQL sequence.
- **trg_decrement_portions_on_order:** After INSERT on public.orders. Atomically parses order items and decrements portionsLeft in public.products.outlets JSONB, automatically switching inStock = false when portions reach 0.
- **handle_new_user:** After INSERT on auth.users. Automatically provisions a public.profiles record with role assignment (owner for admin emails, outlet_manager for hub managers, or customer).

5.3 Security Hardening & Row Level Security (RLS)

All database tables have Row Level Security enabled. Security definer functions get_auth_role() and is_owner() ensure that kitchen managers can never view or modify data outside their assigned outlet_id, and customers can only view their own orders, addresses, and loyalty rewards.

